■■ MIRAGE-SHIELD

Development Roadmap Execution Audit & Pre-Submission Plan

Living-Room Defense Against Digital Arrest & Courier-Extortion Scams

Amazon Build Hackathon 2026 · Ring Primary Track · AWS Builder Track · Open Source Track

● **STATUS: 88% COMPLETE (DAY 4 PRE-FREEZE)**

100% CORE CODE, APIS & ARCHITECTURE VERIFIED

Audit Date: Oct 22, 2026 (08:00 IST)

Target Submission: Oct 23, 2026, 18:00 IST

Hard Deadline: Oct 24, 2026, 00:30 IST

ROADMAP GATE STATUS	AUTOMATED TEST SUITE	MASTER VERIFICATION	SUBMISSION BUFFER
3 of 4 GATES PASSED	**15 / 15 PASSED (100%)**	**3 / 3 STAGES (100%)**	**+6.5 HOURS BUFFER**
Day 1, 2, 3 Passed (Day 4 Active)	Scenarios 1, 2, 3 (5x loops each)	Ring Health, Scenarios, MCP Tests	Target: Oct 23 @ 18:00 IST

1. Executive Summary & Technological Breakthrough

The Threat: In India and globally, 'Digital Arrest' scams have drained hundreds of crores from vulnerable citizens. Impersonating police or customs over video calls, scammers command victims to lock themselves in rooms and threaten immediate arrest: *'Armed officers are outside your front door right now. Transfer ■4,50,000 to the RBI verification escrow account or we break in!'*

The MIRAGE-SHIELD Solution: Scammers win by lying about the physical world while keeping victims trapped on a 6-inch phone. MIRAGE-SHIELD catches the lie using **Ring Video Doorbell Pro** as a physical ground-truth oracle. An autonomous **Bedrock / Strands AI Agent** cross-examines call claims against live Ring sensor events. Upon detecting the physical contradiction (Caller claims police outside vs Ring confirms 0 humans), the agent triggers **4 Living-Room Shields**: (1) **Fire TV Full-Screen Takeover** showing sensor proof and the 1930 helpline, (2) **Ring Chime Siren Alert**, (3) **PaymentGuard Transfer Lock** freezing outgoing bank transfers, and (4) **Family SMS Alert** with secondary co-sign authorization.

2. What Has Been Accomplished: Foundations & Core Pipelines (Days 1 & 2)

Subsystem / Milestone	Roadmap Phase	Engineering Deliverables & Verifiable Evidence	Status
Monorepo & Contracts	Day 1 (Block 1 & 4)	Full monorepo structure created (`ring_service`, `detector`, `agent`, `firetv_app`, `web_console`, `docs`, `shared`, `open_source`). Defined Pydantic schemas (`DoorEvidence`, `CallRisk`, `Verdict`) with offline test fixtures.	**PASSED** 100% Done
Ring Partner API Client	Day 1 & 2 (Pipeline A)	Live HTTPS client (`ring_service/client.py`) calling official `api.amazonvision.com/v1/devices`, events history, and snapshots. Built pre-flight JWT expiry handling and resilient sandbox failover mode.	**PASSED** Live & Verified
Ring Webhook & Chime	Day 2 (Pipeline A)	FastAPI webhook endpoint with HMAC-SHA256 signature verification. Ring Chime acoustic playback triggering `audio_ref_urgent_security_alert` on verified scam detection.	**PASSED** Operational
Scam Detector & NLP	Day 2 (Pipeline B)	Visual frame deepfake detector (`detector/deepfake_detector.py`), STT voice transcription, and Coercion Script Classifier (`detector/nlp_classifier.py`) extracting isolation, warrant, and physical front-door claims.	**PASSED** Operational
Strands / Bedrock AI Brain	Day 2 (Pipeline C)	Cognitive reasoning engine cross-examining call claims against Ring telemetry. Autonomous tool coordination (`get_door_evidence`, `score_call`, `lock_payments`, `trigger_tv_alert`, `notify_family`).	**PASSED** Autonomous
PaymentGuard & Co-Sign	Day 2 (Pipeline C)	Instant programmatic freeze on outgoing UPI/bank transfers (`agent/payment_guard.py`). Rejects fraudulent transfers; unlocks only when family member enters secondary cryptographic co-sign token.	**PASSED** Dual-Control
AWS KMS Digital Ledger	Day 2 (Pipeline C)	Cryptographic digital signing via AWS KMS `RSASSA_PKCS1_V1_5_SHA_256` applied to incident digests combining transcript, Ring snapshot hash, and verdict. Recorded in DynamoDB incident ledger with PITR.	**PASSED** Cryptographic
Fire TV 10-Foot Takeover	Day 2 (Pipeline D)	10-foot living room display application (`firetv_app/`) adhering to Vega OS TV specs (overscan margins, high contrast, D-Pad focus navigation, Ring snapshot evidence panel, 1930 helpline). Sub-5s response.	**PASSED** 10-Foot Compliant

2. What Has Been Accomplished: Integration, Hardening & Verification (Days 3 & 4)

Subsystem / Milestone	Roadmap Phase	Engineering Deliverables & Verifiable Evidence	Status
Master Console Ecosystem	Day 3 (Integration)	Unified interactive web application (`run.py` at `http://127.0.0.1:8000/`) orchestrating Phone View, Fire TV Living Room Takeover, Ring Doorbell Camera, Family Guardian SMS, and AWS Ledger in real time.	**PASSED** **100% Operational**
Open Source MCP Package	Day 3 (Hardening)	Standalone `ring-evidence-mcp` package extracted under MIT License. Exposes `get_door_evidence` and `verify_porch_claim` tools to Claude Desktop, Cursor, and Bedrock agents. 100% standalone test pass.	**PASSED** **PyPI Ready**
Automated 15/15 Scenario Gate	Day 3 Integration Gate	`test_scenarios.py` executes 5 consecutive iterations each of Scenario 1 (Scam Caught), Scenario 2 (Genuine Call), and Scenario 3 (Real Courier). 100% pass rate across all 15 runs.	**PASSED** **15/15 Runs**
Master Verification Suite	Day 4 Pre-Freeze	Unified test runner `verify_all.py` validating: (1) Ring API Health, (2) 15-Run Scenario Suite, and (3) Standalone MCP Protocol server tools. 3/3 stages passing in 16.3 seconds.	**PASSED** **3/3 Stages**

Physical Evidence Oracle: Scenario Visual Assets Verified in Code



Scenario 1: Staged Phone Scam
Claim: 'Police outside your door!'
Deepfake: 0.92 | Coercion: Flagged



Scenario 1: Ring Sensor Oracle
Sensor Truth: Porch 100% CLEAR
0 Humans · 0 Cars · Contradiction Caught!



Scenario 3: Amazon Delivery
Verified Package Delivery
Human + Package Confirmed · Safe

3. Hackathon Track Alignment & Scoring Compliance Matrix

Hackathon Requirement	Rule & Mandatory Criteria	MIRAGE-SHIELD Implementation Status	Compliance
Ring Partner API Calls	Real API calls to `api.amazonvision.com` in main code path.	`ring_service/client.py` makes live HTTPS calls for devices, events, snapshots, and chime audio.	**100% COMPLIANT**
Ring Primary Track	Demonstrate working IoT sensors via device, playground, or emulator.	Both live Ring Developer Playground token and offline sandbox fixture support integrated.	**100% COMPLIANT**
AWS Builder Track	Substantial integration of AWS cloud services with architecture documentation.	AWS KMS asymmetric signing (`RSASSA_PKCS1_V1_5_SHA_256`), Bedrock Claude 3 Haiku, DynamoDB PITR ledger.	**100% COMPLIANT**
Open Source Challenge	Reusable open source package with OSI license and contribution URL.	Standalone `ring-evidence-mcp` (Model Context Protocol) package for Claude Desktop/Cursor/Bedrock.	**100% COMPLIANT**
Friction Log (10% Bonus)	Detailed entries: task, steps, expected vs actual, severity, suggestions.	8 detailed real friction logs documented covering JWT expiry, WHEP SDP, Vega OS WSL, and Chime TTS.	**100% COMPLIANT**
Honest Scoping	Plainly state what is real hardware, simulated software, or test-mode payments.	Declared in README and Devpost submission: Ring (Real), Call (Staged), Payments (Simulated).	**100% COMPLIANT**
Demo Video (< 3 Min)	English, public video hosting, no copyrighted audio, under 3:00.	2:55 script fully written; filming and voice-over scheduled for Oct 22 afternoon.	**PENDING FILM**

4. Roadmap Execution Plan: What Is Yet to Be Done

With all core software, AI models, APIs, and tests verified and frozen (12:00 IST feature freeze), the remaining work centers on **Demo Video Production**, **Devpost Field Finalization**, and **Pre-Submission Review**.

Priority & Time Block	Actionable Task Description	Quality Deliverable & Success Criteria	Target Gate
BLOCK 1: URGENT **Oct 22, Afternoon** **12:00 – 16:30 IST**	**Screen-by-Screen Demo Recording (1080p)** Capture crisp 1080p video clips of all three demo scenarios: • **Scenario 1:** Phone call extortion + Ring oracle check + TV takeover + Payment lock. • **Scenario 2:** Genuine family video call (stays quiet, proving zero false alarms). • **Scenario 3:** Real courier delivery (Amazon package detected, verified delivery). *Record backup takes simultaneously to prevent re-shoots.*	• Raw 1080p MP4 clips for each scenario. • Crisp UI rendering of Fire TV display and Master Console. • Backup take stored in `recordings/` folder.	**IN PROGRESS** **Lead Dev / Video** **Target: 16:30 IST**
BLOCK 2: URGENT **Oct 22, Afternoon** **16:30 – 18:00 IST**	**Voice-Over Narration & Audio Sync** Record voice-over audio following the exact 2:55 roadmap script (see Section 5). • Deliver clear, punchy, authoritative narration. • Deliver key close: *'Scammers lie about the front door. Ring tells the truth.'* • Strictly zero third-party music or copyrighted audio (hackathon rule).	• Clean 48kHz WAV/AAC audio track. • Zero background hiss or room echo. • Exact time-code sync to UI transitions.	**PENDING** **Audio Lead** **Target: 18:00 IST**
BLOCK 3: CRITICAL **Oct 22, Evening** **18:00 – 20:30 IST**	**Video Editing, Timing & YouTube Hosting** Assemble video cuts, titles, split-screen views, and voice-over: • Enforce hard duration constraint: **strictly under 2 minutes 55 seconds**. • Ensure the scam-caught climax occurs inside the **first 60 seconds**. • Export 1080p H.264 video. • Upload to YouTube as **Public**; test mobile playback link.	• Public YouTube URL. • Duration: 2:50 – 2:54. • Verified playable on phone and desktop browsers without login.	**PENDING** **Editor / Team** **Target: 20:30 IST**
BLOCK 4: REVIEW **Oct 22, Night** **20:30 – 22:30 IST**	**Devpost Submission Form Field Populating** Copy and format polished submission texts from repository artifacts: • **Description & Problem:** `docs/SUBMISSION.md` (Executive summary & architecture). • **Friction Log:** `docs/FRICTION_LOG.md` (8 detailed real entries for 10% bonus). • **Product Feedback:** `docs/PRODUCT_FEEDBACK.md` (Ring API, Vega SDK, Bedrock). • **AWS Builder Track:** `docs/AWS_BUILDER.md` (KMS, Bedrock, DynamoDB). • **Honest Scoping:** Declare Real vs Simulated vs Emulated clearly.	• All Devpost fields drafted and saved in Devpost portal as draft. • Markdown formatting rendered without broken tags.	**READY** **Documentation Lead** **Target: 22:30 IST**
BLOCK 5: PRE-FLIGHT **Oct 23, Morning** **10:00 – 14:00 IST**	**Public Repository Setup & Clean Clone Audit** • Ensure GitHub repository is set to **Public** with standard MIT License. • Remove any temporary test files or local caches (`__pycache__`, `.env` secrets). • Perform a clean-clone test on a secondary machine following `README.md` verbatim. • Verify all clickable links in documentation resolve properly.	• Public GitHub repository link. • One-command `python run.py` works seamlessly on fresh machine. • No sensitive credentials committed.	**SCHEDULED** **QA / DevOps** **Target: 14:00 IST**
BLOCK 6: SUBMISSION **Oct 23, Afternoon** **14:00 – 18:00 IST** ***(6.5h Before Cutoff)***	**Two-Person Sign-Off & Devpost Final Submission** • Two team members perform a comprehensive line-by-line read of every submission field. • Confirm track selection: Primary (Ring), Mini-Challenges (AWS Builder & Open Source). • Hit **'Submit Project'** on Devpost before **18:00 IST**. • Capture full-page screenshot of submission confirmation screen.	• Verified Devpost project submission receipt. • Confirmation screenshot archived. • Zero risk of last-minute portal outages.	**FINAL GATE** **Full Team Sign-Off** **Target: 18:00 IST**

5. Demo Video Production Blueprint & Timing Script (2 Minutes 55 Seconds)

Judges evaluate projects in under 3 minutes; the core breakthrough must land inside the first 60 seconds. The video will be filmed and edited strictly according to this timeline:

Timestamp	On-Screen Visual Action	Voice-Over Narration Script (Exact Delivery)	Scene Goal
0:00 – 0:15 (15 sec)	Black screen with bold statistic: ■*120+ Crores lost to Digital Arrest scams*. Transition to phone ringing with high urgency.	*'Digital arrest scams are draining innocent citizens worldwide. Scammers trap victims in psychological terror by telling one devastating lie about the physical world.'*	Hook & Urgency
0:15 – 0:55 (40 sec)	Split-screen: Fake police officer in uniform demanding ■4.5L escrow. Living room TV and Ring live feed shown on right.	*'The fake officer claims: "Armed police are outside your door right now. Transfer the verification escrow immediately or we break in!" On the phone, deepfake score climbs to 0.92; coercion script flags isolation.'*	The Physical Lie
0:55 – 1:25 (30 sec)	**THE CLIMAX:** Strands Agent checks Ring sensor oracle. Porch is CLEAR. Fire TV takes over full screen in high-contrast alert. Chime sounds. Bank transfer blocked.	*'MIRAGE-SHIELD's AI agent checks the Ring Doorbell oracle. Zero people. Zero vehicles. It is a physical contradiction! Instantly, Fire TV takes over the living room, shattering the isolation. The chime rings. Bank transfers freeze.'*	**Living-Room Takeover**
1:25 – 1:45 (20 sec)	Family phone shows SMS alert with verified snapshot hash. Son reviews incident and enters secondary co-sign authorization.	*'Simultaneously, trusted family receives an emergency SMS alert with cryptographic proof. Money remains locked until a family guardian co-signs to release it.'*	Dual Co-Sign Guard
1:45 – 2:10 (25 sec)	Rapid sequence: Scenario 2 (genuine daughter call - stays quiet) and Scenario 3 (real Amazon delivery - package and human verified).	*'MIRAGE-SHIELD never cries wolf. In genuine family calls, it remains silent. When a real Amazon courier arrives, Ring confirms package and human events, permitting normal activity.'*	Zero False Alarms
2:10 – 2:35 (25 sec)	Architecture diagram highlighting Ring API endpoints, Bedrock Strands agent, and AWS KMS cryptographic signature ledger (`RSASSA_PKCS1_V1_5_SHA_256`).	*'Built on official Ring Partner APIs, Amazon Bedrock, and AWS KMS for tamper-evident legal records admissible under cybercrime reporting portal 1930.'*	AWS Architecture
2:35 – 2:55 (20 sec)	Open-source `ring-evidence-mcp` package repository, team credits, and closing logo banner.	*'We have also open-sourced ring-evidence-mcp so any AI agent can verify physical entrance truth. Scammers weaponize isolation and lie about the front door. Ring tells the truth. MIRAGE-SHIELD defends the living room.'*	Open Source & Close

6. Risk Contingency & Approved Graceful Cut Order

If video rendering or submission window compresses before the Oct 23, 18:00 IST submission deadline, the team must strictly follow the roadmap's approved graceful cut order:

• **Approved Cut Order:** (1) *Chime audio playback* (Fire TV carries warning) ➔ (2) *Family co-sign UI screen* (Keep SMS text and payment lock) ➔ (3) *AgentCore deployment* (Utilize verified Lambda / local runtime) ➔ (4) *Scenario 3 Courier footage* (Focus on Scenario 1 & 2) ➔ (5) *Deepfake video model score* (Rely on NLP coercion transcript analysis + Ring physical contradiction).

• **NON-NEGOTIABLE CORE (NEVER CUT UNDER ANY CIRCUMSTANCES): ✔ Real Ring API Calls** (`api.amazonvision.com` device ingestion & snapshot hash verification) | ✔ **Fire TV Full-Screen Intervention Takeover** (The 10-foot visual proof shattering the scam isolation) | ✔ **Scenario 1 (Scam Caught)** (The primary digital arrest cross-examination demonstration).

7. Final Pre-Flight Submission Checklist (Target: Oct 23 @ 18:00 IST)

Item	Submission Requirement & Verification Step	Target Deadline	Status
1	Demo Video uploaded to YouTube/Vimeo as Public (Under 2:55, English, no 3rd-party audio)	Oct 22, 20:30 IST	**IN PROGRESS**
2	Devpost description text, architecture diagram, and honest scoping declarations pasted	Oct 22, 22:30 IST	**DRAFT READY**
3	Friction Log (8 entries) and Product Feedback (Ring, Vega, Bedrock, Kiro) populated	Oct 22, 22:30 IST	**DRAFT READY**
4	AWS Builder fields and Open Source (`ring-evidence-mcp`) links and descriptions verified	Oct 22, 22:30 IST	**DRAFT READY**
5	GitHub repository toggled to Public with MIT License (`LICENSE` verified)	Oct 23, 12:00 IST	**SCHEDULED**
6	Final two-person review of all Devpost submission form fields	Oct 23, 16:00 IST	**SCHEDULED**
7	Official Devpost submission finalized before 18:00 IST (+6.5 hour buffer) & screenshot saved	Oct 23, 18:00 IST	**TARGET GATE**

8. Technical Verification Sign-Off & Audit Certificate

MASTER PRE-FREEZE VERIFICATION CERTIFICATE

- **Master Test Runner:** `python verify_all.py` executed cleanly with **3 of 3 stages passing** in 16.3s.
- **Integration Scenarios:** `python test_scenarios.py` completed **15 of 15 consecutive iterations (100% pass rate)** across Scenarios 1, 2, and 3.
- **Open Source MCP Server:** `ring-evidence-mcp` standalone test suite passed **4 of 4 tool registration and protocol dispatch tests**.
- **Cryptographic Chain of Custody:** AWS KMS asymmetric signature generation verified (`RSASSA_PKCS1_V1_5_SHA_256`).
- **Code Freeze Status: FROZEN as of Oct 22, 12:00 IST**. Zero architectural or breaking code changes permitted during video production.

9. Project Documentation Artifacts & Resource Index

Document Artifact	File Path in Workspace	Purpose & Hackathon Deliverable Mapping
Full Submission Package	`docs/SUBMISSION.md`	Complete 320+ line Devpost submission draft with executive summary, friction log, and honest scoping.
Developer Friction Log	`docs/FRICTION_LOG.md`	8 detailed developer friction entries across Ring API, Vega OS, and AWS KMS (10% bonus submission).
Product Feedback	`docs/PRODUCT_FEEDBACK.md`	Structured actionable feedback for Ring API, Vega SDK, Bedrock, and Kiro product teams.
AWS Builder Architecture	`docs/AWS_BUILDER.md`	Technical reference for Bedrock Strands Agent, KMS asymmetric signing, and DynamoDB incident ledger.
Open Source MCP Server	`ring-evidence-mcp/`	Standalone Model Context Protocol server exposing Ring sensor tools to AI agents under MIT license.
Unified Server Launcher	`run.py`	Single-command master launcher hosting Master Console, Fire TV view, and API docs at `127.0.0.1:8000`.